Tropical Cyclone Threat Days Elevate All-Cause Mortality Before Landfall in Florida

James B. Elsner, Christopher K. Uejio*, David Hsu, et al.

Abstract

Tropical cyclones are known to cause deaths at and after landfall, but the timing of broader mortality impacts remains unclear. We quantify short-term, day-specific changes in all-cause mortality surrounding tropical cyclone exposure across Florida using geocoded death records, modeled storm swaths, and a multi-decade ZIP-code panel. Fixed-effects Poisson models compare mortality on days before and after impact with non-storm days, controlling for spatial and temporal confounders. Mortality increases consistently during the pre-impact threat period, not only at landfall and during recovery. For storms with sustained winds ≥ 33 m/s (≥ 64 kt), mortality is elevated by ~9–11% on several pre-impact days and rises further at impact and the following day. Weaker storms show the same pattern with smaller magnitudes, and two placebo designs yield no comparable pre-impact elevation. The strongest effects occur for older adults, and cause-specific analyses show post-impact injury spikes. These findings indicate that cyclone-related mortality begins before landfall, underscoring the need to account for threat-phase risks in preparedness policy.

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Introduction

Tropical cyclones (TCs) contribute to mortality through both direct and indirect mechanisms. Direct deaths (e.g., drowning, trauma, electrocution) most often occur during landfall and immediate impact, whereas indirect deaths arise through diffuse pathways including interruption of medical care, power outages affecting climate control and medical devices, carbon monoxide poisoning, infectious/environmental exposures, and delayed emergency response. Post-event surveillance and attribution studies consistently find that indirect deaths account for a large share of storm-associated mortality and can equal or exceed direct counts in major events, indicating that official storm fatality totals substantially understate the total mortality burden (1-4).

These indirect pathways are time-distributed. Health risks may begin before landfall, intensify during impact, and persist throughout recovery. However, many operational reporting systems and empirical analyses emphasize deaths occurring at or after impact, leaving the temporal onset of excess mortality incompletely characterized (4-6). A common framework for quantifying storm mortality is excess mortality analysis, wherein observed deaths during/after exposure are compared to an expected baseline adjusted for seasonality and trends. Time-series and panel Poisson models, negative binomial models, and Bayesian hierarchical approaches are widely used to estimate relative risks associated with storm windows (4-6). Recent multi-storm analyses at county-to-national scales report modest but statistically significant increases in mortality following cyclone exposure across multiple causes (4-6). Meta-analyses similarly estimate short-term mortality increases on the order of five to ten percent after TC events, with substantial heterogeneity across regions and storm characteristics (5). Longer-horizon analyses suggest mortality effects can persist for months to years after major storms (6).

Methodological advances in environmental and disaster epidemiology emphasize careful control for seasonality, long-term trends, and spatial heterogeneity when estimating excess mortality (7, 8). Fixed-effects panel designs and time-stratified models help isolate short-term shocks from background variation, while distributed-lag frameworks allow effects to vary across event time (8, 9). Despite these advances, relatively few studies explicitly isolate mortality risk in the days preceding impact; most define exposure windows beginning at landfall or shortly thereafter, and

when pre-event days are included, their coefficients are rarely foregrounded or interpreted (4-6). As a consequence, empirical evidence on threat-phase mortality remains limited.

Pre-impact risk is nevertheless plausible. Evacuation orders, transport disruptions, pharmacy closures, dialysis interruptions, electricity contingency planning, and acute stress responses often occur before landfall; vulnerable populations (e.g., older adults, socially isolated individuals) may face heightened risk during preparation phases (1, 3, 10). A further limitation of prior work is spatial resolution: many studies rely on county-level units, potentially obscuring within-region heterogeneity. Finer spatial units enable better alignment between storm swaths and exposed populations and improve control for local baseline mortality differences. Recent geospatial work documents spatial clustering in storm-related health impacts, but high-resolution event-time mortality analyses remain uncommon (11, 12).

Here we combine point-level mortality data, storm-swath exposure modeling, and high-resolution fixed-effects panel methods to estimate day-specific event-time mortality profiles around hurricane impact in Florida. Instead of aggregating all pre- or post-storm days, we estimate rate ratios for each day from −8 to +8, evaluate consistency across demographic and cause-specific subsets, and assess intensity dependence. We find that mortality risk measurably increases before storm impact, reframing TC mortality as a temporally distributed phenomenon with implications for warning systems, preparedness policy, and climate adaptation planning. Our findings complement and extend recent multi-storm analyses and excess mortality frameworks by clarifying when risk *emerges* relative to impact (4-9, 12).

Results

Study population, exposure, and design

We assembled a statewide, ZIP-code-level daily panel (8,144,840 ZIP-day observations) linking geocoded deaths to modeled TC swath exposure (Figure 1). Event time was defined relative to each storm's Florida impact date (Date0), spanning −8 through +8 days. Our primary specification is a Poisson fixed-effects (FE) event-study with the following high-dimensional fixed effects: ZIP, year×week-of-year (yw), day of week (dow), and a ZIP×COVID-period effect (January 1, 2020–December 31, 2022). Standard errors are two-way clustered by ZIP and year-week. For the hurricane model (maximum sustained 10-m winds of ≥ 33 m/s [≥ 64 kt]), we estimate 1,006 ZIP FEs, 1,195 year-week FEs, 7 day-of-week FEs, and 1,988 ZIP×COVID FEs. A small number of fixed-effect strata (<5%) with either all-zero outcomes or only one usable observation were automatically dropped, as such strata do not contribute to identification in a fixed-effects Poisson model.

Event-time mortality profile (hurricanes [≥ 33 m/s])

The all-cause mortality profile exhibits multi-day elevations before landfall, pronounced peaks at and immediately after impact, and a gradual decline during the cleanup period (Figure 2). Pre-impact days show repeated positive departures from baseline with clustering across adjacent days: −8 RR=1.09, −7 RR=1.09, −4 RR=1.07, −3 RR=1.10, and −1 RR=1.11. At impact, Date0 RR=1.15, with a peak on the following day (+1 RR=1.21). Elevated risk persists through the first post-impact week (+2 RR=1.17; +3 RR=1.09; +4 RR=1.06). Several coefficients are statistically significant at conventional levels, and the temporal coherence indicates a structured event-time response rather than isolated spikes.

Intensity stratification

Restricting to ≥28 m/s storms preserves the shape but with smaller magnitudes (Figure 3); pre-impact days are typically in the 5–10% range (−8 RR=1.10; −7 RR=1.05; −6 RR=1.05; −5 RR=1.08; −4 RR=1.06; −1 RR=1.10), with Date0 RR=1.07 and +1…+2 around 1.08–1.09. Including ≥17 m/s storms (tropical storms) yields a smaller but still coherent pattern (−8 RR=1.05; −7 RR=1.06; −5 RR=1.06; −3 RR=1.06; −1 RR=1.07; Date0 RR=1.05; +1 RR=1.06;

+2 RR=1.06). Thus, pre-impact risk scales with intensity while maintaining the same temporal structure.

Identification and Falsification Tests

To assess whether the pre-impact elevation reflects residual seasonal timing or model specification rather than storm exposure, we conducted three complementary checks. First, a temporal permutation placebo reassigning storm impact dates within the hurricane season (±16-day exclusion from the true Date0) yields no coherent pre-impact pattern. Coefficients before pseudo-Date0 are centered near the null and do not exhibit multi-day clustering (Figure S1).

Second, we implemented a spatial permutation falsification that preserves the exact daily exposure counts and event-time structure while breaking geographic alignment. Within each calendar date, event-time labels were randomly reassigned across ZIP codes. Across 200 randomized exposure calendars, the observed pre-impact window average (−8 through −1) rate ratio of 1.067 exceeded all placebo estimates; the largest placebo RR was 1.040 (empirical one-sided $p < 0.005$). No placebo draw produced a coherent multi-day pre-impact run comparable to the observed pattern (Figure S2). Because this permutation preserves calendar-time composition while eliminating spatial correspondence between storm swaths and local populations, the result indicates that the threat-phase elevation depends on correct geographic alignment.

Third, we extended the event-time window to −16 days to assess potential pre-trend bias. The far-pre window (−16 through −9) shows a smaller average increase (RR = 1.04, 95% CI: 1.00–1.08) than the near-pre window (−8 through −1; RR = 1.07, 95% CI: 1.03–1.10), with the largest effects occurring at and immediately after impact. This graded escalation toward landfall is inconsistent with a stable differential pre-trend. Although isolated far-pre days attain statistical significance, they do not form sustained runs and are smaller in magnitude than near-impact effects. Together, these checks indicate that the pre-impact elevation is not an artifact of seasonal timing, fixed-effects structure, or mechanical properties of the exposure calendar.

Demographic subsets

Age-restricted models (≥ 60 years) show larger and clearer pre-impact elevations than the pooled population: −8 RR=1.11; −4 RR=1.08; −3 RR=1.14; −1 RR=1.13, followed by Date0 RR=1.19,

+1 RR=1.22, and +2 RR=1.20. Male-only and white-only subsets broadly mirror the main pattern, with stronger post-impact peaks (e.g., men: −7 RR=1.13; 0 RR=1.20; +1 RR=1.17; white: −3 RR=1.11; −1 RR=1.10; 0 RR=1.17; +1 RR=1.22; +2 RR=1.17). The married-only subset shows greater variability and wider intervals, consistent with smaller daily counts; still, it shows Date0 RR=1.16 and +1 RR=1.21.

Cause-specific analyses

Cause-specific timing is heterogeneous. Injury deaths show minimal or negative pre-impact movement (e.g., −1 RR=0.82) but large post-impact spikes: Date0 RR=1.77, +1 RR=1.32, +2 RR=1.95, with additional elevations on +4 RR=1.43 and +6 RR=1.46. Cardiovascular mortality exhibits distributed elevations spanning both pre- and post-impact days (e.g., −6 RR=1.34; −3 RR=1.47; −1 RR=1.16; +1 RR=1.41; +5 RR=1.25; +6 RR=1.38), consistent with stress and care-access pathways. Respiratory and neuropsychiatric estimates are less stable due to sparse counts and estimator warnings in some specifications; we treat these as exploratory.

Pre- versus post-impact windows

Averaging across pre-impact (−8 through −1) and post-impact (+1 through +8) windows in the hurricane model indicates similar magnitudes with overlapping uncertainty, suggesting that excess mortality is not confined solely to the cleanup period. Because coefficients are estimated jointly within a single event-time model, the window contrasts reflect a continuous response profile rather than independent tests.

Discussion

This study provides evidence that tropical cyclone mortality risk is temporally distributed and begins before landfall. Using high-resolution geocoded mortality data linked to modeled storm exposure and analyzed in a fixed-effects event-time framework, we detect multi-day elevations in all-cause mortality during the threat phase that persist through impact and cleanup. For hurricanes (≥ 33 m/s), pre-impact elevations of ~9–11% on several days precede a ~15% increase at impact and a ~21% increase the following day; weaker storms display the same shape with smaller magnitudes. A placebo event-time analysis shows no comparable pre-impact elevation, supporting internal validity.

Mechanistically, the timing of injury mortality with minimal pre-impact and spiking after impact is consistent with trauma, debris cleanup, generator/carbon monoxide risks, and mobility hazards. By contrast, all-cause and cardiovascular mortality exhibit distributed elevations both before and after impact, aligning with anticipatory stress, healthcare access disruptions (e.g., pharmacy closures, dialysis interruptions), and interdependent lifeline system failures that begin during the forecast/warning period. The older-adult risk profile featuring larger, clearer elevations from pre-impact into early post-impact highlights the need for targeted continuity-of-care and electricity-dependent therapy support in preparedness operations.

The coherent clustering of elevated coefficients across adjacent pre-impact days, combined with intensity scaling (effect sizes are larger for stronger storms), argues for a structured population-level response rather than isolated anomalies or seasonal confounding. The distributed shape also argues against short-term mortality displacement; we do not observe compensating deficits after the pre-impact cluster. An extended-lead specification does not show a flat zero pre-period; however, the effect size increases systematically as impact approaches. This temporal gradient argues against a stable differential pre-trend and supports interpretation as anticipatory risk rather than baseline confounding.

Implications follow for preparedness and adaptation. If mortality risk rises during the threat phase, preparedness should include proactive continuity-of-care strategies including advance prescription refills, dialysis scheduling and transport, backup power for medical devices, prioritized emergency transport, and targeted outreach to high-risk populations, rather than

focusing exclusively on impact mitigation. From a measurement standpoint, attribution studies and excess mortality frameworks that start exposure windows at landfall likely underestimate total mortality burden (4-7). Finally, as climate change increases the proportion of high-intensity TCs, the burden associated with both anticipatory and recovery phases may grow even if overall frequency remains stable (13).

Limitations include possible residual confounding from unobserved, time-varying factors; exposure defined via spatial intersection with modeled swaths (not capturing evacuation); and small daily counts in cause-specific strata leading to wider uncertainty and estimator warnings. Estimates describe associations rather than causal effects.

Across multiple specifications and strata, we find that pre-impact days carry measurable mortality risk. Reframing TC risk as a temporally distributed process spanning threat, impact, and recovery can help align public health preparedness and climate adaptation with intensifying storm hazards.

Methods

Study setting and period

We examined short-term changes in all-cause mortality surrounding TC exposure in Florida over 1985–2022, restricting models to the hurricane season (May–November). Analyses were conducted at the ZCTA×day level. The event-time window spanned −8 through +8 days relative to each storm's impact date (Date0).

Storm track data and exposure modeling

We used IBTrACS North Atlantic lines shapefile (v04r00) as the source for TC tracks and metadata (date–time, location, agency wind, pressure, radius of maximum winds). Tracks were read and transformed to EPSG:32616 (UTM Zone 16N) for buffering. We filtered records to the study period and retained storm-hours where wind met intensity thresholds: primary ≥ 33 m/s (≥ 64 kt); robustness ≥ 28 m/s (≥ 55 kt) and ≥ 17 m/s (≥ 34 kt). Missing values for radius-to-maximum winds were imputed through within-storm mean, then by mean within the same central pressure stratum, and re-checked within storm. Track segments were buffered by radius-to-maximum winds to create hourly swaths, intersected with Florida, and aggregated by storm-day. Date0 was the first Florida-intersecting date per storm; we recorded maximum wind over Florida and assigned the Saffir–Simpson category for description. Geometries were transformed to EPSG:4326 for joins.

Event-time polygons (threat-phase calendar)

For each storm, we expanded Date0 to a relative-day grid (−8 through +8) and assigned Date = Date0 + rel_day. For every relative day, the storm's Florida swath geometry was carried forward, creating a time-indexed set of TC polygons. When multiple storms overlapped on a given date, we retained the closest-to-impact assignment; ties were broken in favor of pre-impact.

ZIP boundaries and exposure calendar

We used 2020 ZCTA polygons (tigris cartographic boundaries). After transforming to a common CRS, we intersected Florida ZCTAs with each day's storm swath to identify ZIP×date pairs within the modeled threat envelope. We created a complete ZIP×date grid across study dates; for exposed dates, rel_day was populated and rel_day_f took values in {None, −8 through +8}. If a

ZIP×date intersected more than one storm, we kept the smallest |rel_day|, tie-breaking to pre-impact.

Mortality data, geocoding, and aggregation

We assembled individual-level death records for Florida residents (all causes), geocoded final residence coordinates (WGS84), and spatially joined deaths to ZCTAs (largest-area match when needed). We filtered to study dates and removed deaths lacking spatial coordinates or failing ZIP assignment. For all-cause analysis, the dataset contained approximately 5.85 million deaths. We aggregated to daily counts per ZIP and left-joined the ZIP×date exposure calendar to produce a panel with one row per ZIP×date containing the outcome (deaths) and rel_day_f.

Statistical analysis

Our primary specification estimates event-time rate ratios using Poisson pseudo-maximum likelihood with high-dimensional fixed effects and two-way clustered inference. Predictors: categorical rel_day_f (−8 through +8; reference = None). Fixed effects: ZIP, year×week-of-year, day-of-week, and ZIP×COVID (January 1, 2020–December 31, 2022). Inference uses two-way clustered sandwich SEs (clusters = ZIP and year-week). We report RR = exp(β) and 95% CIs and emphasize temporal structure (coherent runs of positive coefficients) rather than isolated p-values. In sparse strata (e.g., cause-specific models), empty event-time levels were dropped to avoid near-singularity; where the cluster-robust variance–covariance matrix was not positive semidefinite, indicating unstable inference, results are interpreted as exploratory.

Model diagnostics for ≥33 m/s, all-cause: Observations = 8,144,840; fixed effects—ZIP = 1,006; year-week = 1,195; day-of-week = 7; ZIP×COVID = 1,988; two-way clustered SE; 40/0/0/104 FE strata (361,232 obs) removed due to all zeros/singletons. In the extended-lead model, the two-way clustered variance estimator required numerical adjustment to ensure positive semi-definiteness. Point estimates were unaffected; inference is therefore interpreted cautiously, and emphasis is placed on window-averaged effects and temporal structure rather than isolated p-values.

Falsification designs

To assess whether pre-impact elevations could arise from seasonal structure alone, we randomized Date0 within the same hurricane season/year for each storm, enforcing a ±16-day buffer from the true impact Date0. We regenerated the exposure calendar and fit the identical

Poisson FE model. Placebo estimates showed no coherent pre-impact elevations, supporting the internal validity of the threat-phase pattern.

To assess whether pre-impact elevations could arise from spatial structure, we randomized the ZIP assignments of the exposure calendar within date (and within coarse geographic strata), preserving the number of exposed ZIPs and the distribution of event-time indicators each day while breaking the spatial correspondence to storm hazards. The randomized exposure produces no coherent pre-impact elevation.

Subgroup, cause-specific, and intensity analyses

We re-estimated the main model for demographic subgroups (e.g., age ≥ 60, sex-restricted, race-restricted, marital status) and for cause-specific categories (e.g., cardiovascular, respiratory, injury, neuropsychiatric). We also restricted exposure to ≥ 17 m/s (≥ 34 kt) and ≥ 28 m/s (≥ 55 kt) thresholds to assess intensity scaling.

Sensitivity to wind-field definition

The primary exposure definition buffers storm track segments by the radius of maximum winds, representing the core wind field. To assess sensitivity to exposure geometry, we expanded the wind field to approximately 2.5×radius of maximum winds, approximating the gale-force radii.

Under the larger exposure definition, pre-impact coefficients attenuate in magnitude and are less consistently significant, whereas post-impact elevations remain evident (e.g., +1 and +2 days). Model fit statistics remain nearly identical to the primary specification.

This pattern is consistent with classical exposure misclassification: expanding the swath increases the number of ZIP-day observations classified as exposed, diluting estimated effects. Importantly, the persistence of post-impact elevations and the retention of temporal clustering near impact suggest that the primary findings are not artifacts of an overly restrictive wind-field definition.

Software

Analyses were conducted in R (≥ 4.2). Spatial processing used **sf**, **dplyr**, **data.table**, **lubridate**, **tidyr**, **USAboundaries**, and **tigris**. Statistical modeling used **fixest** (Poisson PML with

high-dimensional fixed effects and cluster-robust SE) and **broom** for tidying outputs. Spatial joins were chunked for memory efficiency.

Ethics and data security

Individual-level death records were handled under data-use agreements and de-identified prior to analysis. All spatial joins and aggregations were performed within a secure environment and in conformance with applicable institutional review requirements.

Reproducibility and data availability

IBTrACS storm tracks (North Atlantic, v04r00) and U.S. Census/tigris boundary files are publicly available. Analysis code follows the workflow described here; scripts generate storm-day polygons, the ZIP×date exposure calendar, the daily deaths panel, and the Poisson FE models with clustered inference. Access to individual-level mortality microdata is subject to data-use restrictions.

Figure captions

Figure 1. Constructing storm exposure and event time. Panel A: Example storm track with buffered radius to maximum winds-based swath intersecting Florida. Panel B: ZIP polygons within Hurricane Michael's swath. Panel C: ZIP×day exposure calendar indicating relative-day assignments (−8 through +8). Exposure was defined by buffering IBTrACS track segments by the radius of maximum winds, generating daily swaths intersecting Florida. Each storm-day was expanded into relative event time.

Figure 2. Event-time and averaged pre/post hurricane rate ratios for daily mortality. Panel A: Event-study estimates (RRs and 95% CIs) across −8 through +8 (reference = non-storm days). The curve shows pre-impact elevations for hurricanes (e.g., −8 ≈ 1.09; −7 ≈ 1.09; −3 ≈ 1.10; −1 ≈ 1.11), with peaks at Date0 (≈ 1.15) and +1 (≈ 1.21). Panel B: Average pre-impact (−8 through −1) and post-impact (+1 through +8) RRs with 95% CIs; window averages are similar with overlapping uncertainty.

Figure 3. Pre-impact average RRs by intensity threshold. Estimated average RR (days −8 through −1) for ≥ 17 m/s, ≥ 28 m/s, and ≥ 33 m/s storms. Pre-impact averages increase with intensity while preserving the same temporal shape (threat-phase onset).

Figure S1. Main vs. placebo pre-impact estimates. Daily RRs for the pre-impact window (−8 through −1) are shown for the main hurricane model (filled points) and placebo model (open points). The main model displays coherent pre-impact elevations; placebo estimates are near the null with no clustering, supporting the absence of pre-trend bias.

Figure S2. Spatial permutation falsification test. Histogram of pre-impact window (−8 through −1) rate ratios from 200 spatially permuted exposure calendars. Within each date, event-time labels were randomly reassigned across ZIP codes, preserving daily exposure counts while breaking geographic alignment with storm swaths. The vertical red line marks the observed pre-impact RR (1.067). The largest placebo RR was 1.040; the observed estimate exceeded all 200 draws (empirical one-sided $p = 0.005$).

Supplement

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Table S1. Selected Daily Rate Ratios (RRs) for Key Models. RR = exp(β); reference = non-storm days ("None"). Values rounded to two decimals for readability.

Model / Threshold	−8	−7	−3	−1	0 (Date0)	+1	+2
Hurricanes ≥ 33 m/s	1.09	1.09	1.10	1.11	1.15	1.21	1.17
Storms ≥ 28 m/s	1.10	1.05	1.04	1.10	1.07	1.08	1.09
Storms ≥ 17 m/s	1.05	1.06	1.06	1.07	1.05	1.06	1.06
Age ≥ 60 years	1.11	1.09	1.14	1.13	1.19	1.22	1.20
Injury deaths	0.91	1.14	0.88	0.82	1.77	1.32	1.95

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Author contributions

Conceptualization: CU, JBE; Data curation: DH, JBE; Formal analysis: JBE; Investigation: JBE; Methodology: JBE, CU; Visualization: JBE; Writing—original draft: JBE; Writing—review & editing: JBE

Competing interests

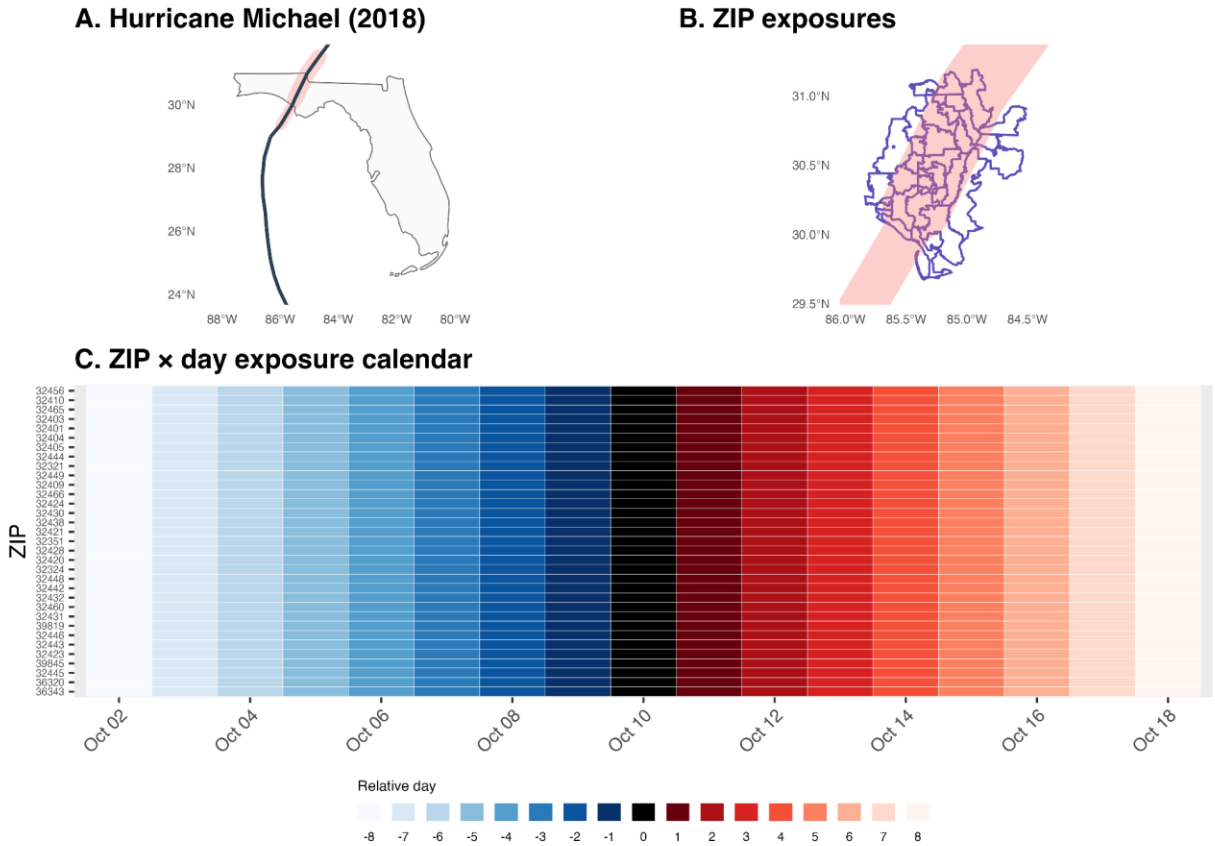
The authors declare no competing interests.

Data availability

IBTrACS storm tracks (North Atlantic, v04r00) and boundary files are publicly available. Access to individual-level mortality microdata is restricted by data-use agreements. Aggregated outputs and code to reproduce the analysis are available upon reasonable request.

Code availability

Analysis code following the workflow described in the Methods is available at https://github.com/jelsner/HO-storms



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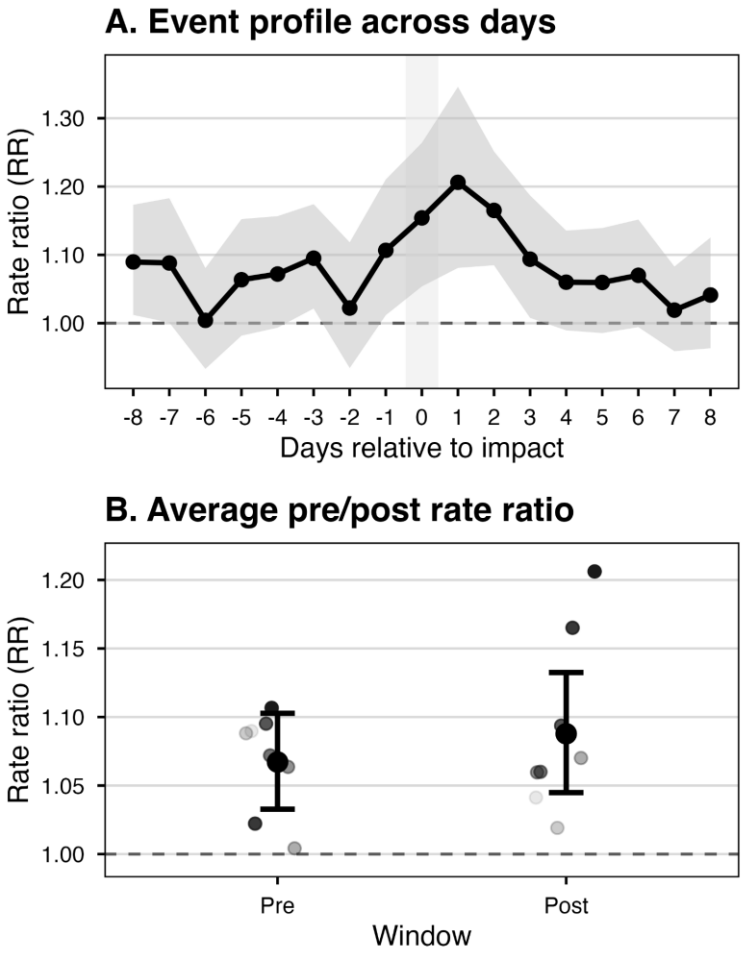

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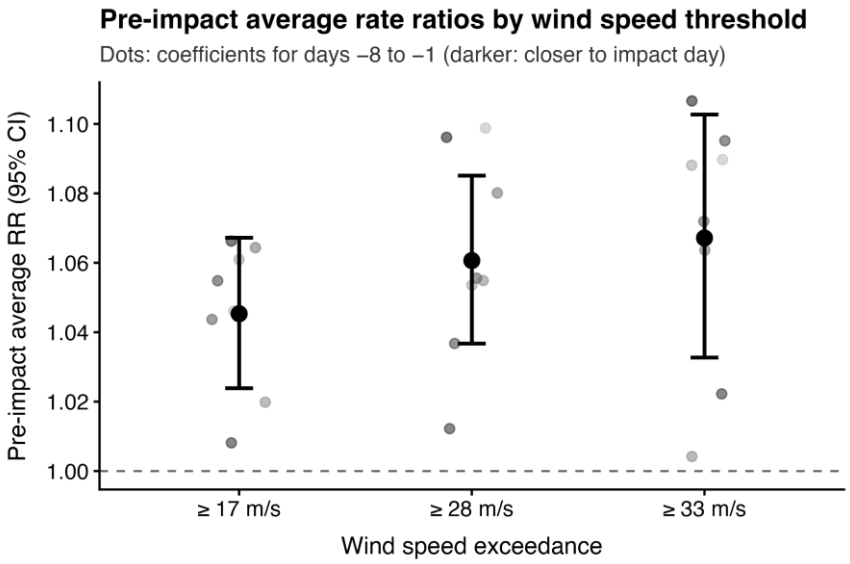

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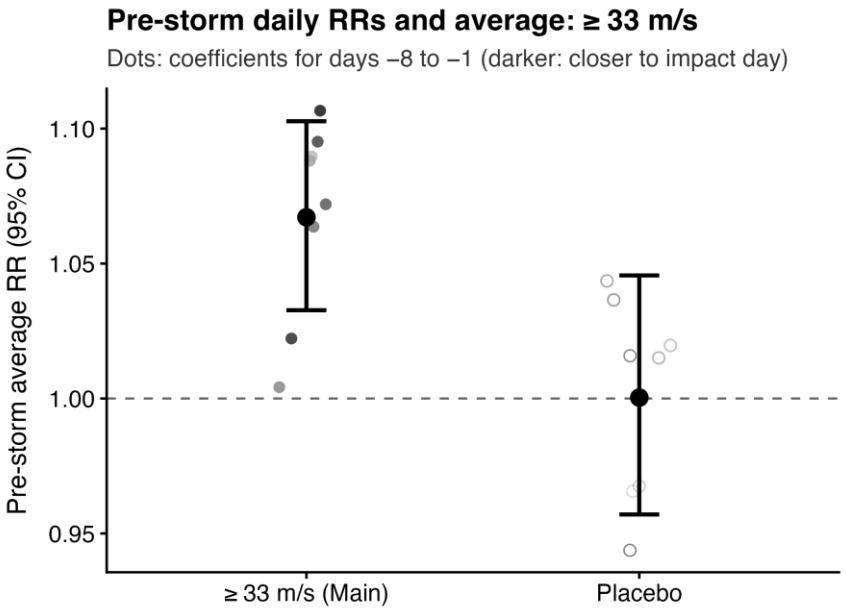

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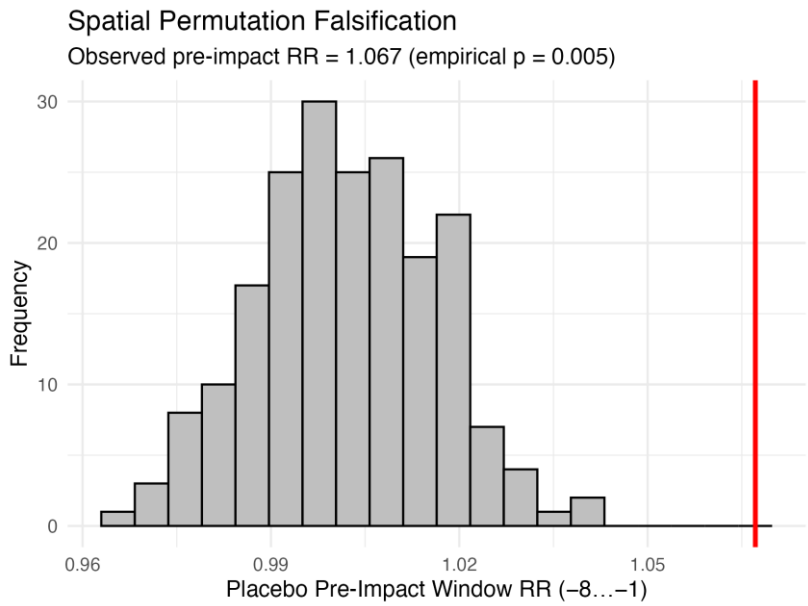

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